BandLink system architecture

Every device in the system and the software on each: two bracelets running the same STM32U5G9 FreeRTOS firmware, the radio links between them, the optional iPhone app over BLE, and the dock and development PC. Solid links are in the MVP or Stretch 1; dashed links are Stretch 2 or optional. This shows the final (490B) bracelet; the 490A prototype differences are listed below. Design in progress: no firmware is written yet.

GNSS satellites

GPS · Galileo · BeiDou · GLONASS (L1, receive only)

L1 signals

Bracelet A

STM32U5G9 · FreeRTOS

Application
Finder · Group · Pairing · Watch face
Mode controller (Standby / Typical / Finding)

UI · LVGL 9
Widgets, encoder + touch input,
partial render, display flush (TE sync)

Services
Link (TDMA, relay S2) · Crypto AES-CCM · Nav
Pairing · Ranging S2 · Battery · Haptics · Phone link OPT-07

RTOS · FreeRTOS
7 tasks · notifications · queues · mutexes
tickless idle → Stop 2 · power manager · flash store

Drivers
SX1262 · MAX-M10S · PN7150 · BNO085 · CO5300
DW3000 · DRV2605L · MAX17048 · BQ25180 · PEC06

HAL / BSP
STM32CubeU5: SPI1–3, I2C1–2, USART, OCTOSPI,
GPDMA, TIM, EXTI, AES/RNG, GPU2D

Hardware
MCU, 500 mAh LiPo, BQ25180 / TPS63802 / MAX17048,
AMOLED, PEC06 crown, DRV2605L + LRA

GNSS MAX-M10S

LoRa SX1262

NFC PN7150

UWB S2 DWM3000

BLE OPT-07 nRF52 bridge
UART link

pogo UART + VBUS · SWD

Dock + development PC

Pogo dock	USART1 → ST system bootloader, debug console; VBUS charging through BQ25180
Dev PC	CubeMX / CubeIDE, on-board ST-LINK (SWD), PPK2
GitHub	Team repo, branches, PR reviews

LoRa 915 MHz
AES-128-CCM position packets · GPS-timed slots

NFC tap (< 10 cm)
One-time pairing: group key + device ID

UWB DS-TWR (S2)
Distance ±0.3 m within ~30 m · 10 Hz in Finding mode

BLE GATT (OPT-07)
Peers · Self status ·
Command · Config
LESC pairing, 6-digit
code on bracelet screen
Group key never sent

Bracelet B

paired friend · same firmware

GNSS MAX-M10S

LoRa SX1262

NFC PN7150

UWB S2 DWM3000

Same firmware stack as Bracelet A

Application · UI · Services · FreeRTOS
Drivers · HAL / BSP · Hardware

Holds the same group key after the NFC tap
Decrypts A's packets, shows A's arrow + distance
Sends its own position in its TDMA slot

Relay (S2): re-broadcasts A's packet
with hop count + 1 so a third bracelet
out of A's range still hears it

iPhone app (OPT-07)

Swift · SwiftUI · iOS

Views	Radar (works offline) · Map · Group Messages · Settings
App logic	BLEManager: Core Bluetooth, background mode, state restoration · BraceletClient: GATT decode
Platform	Core Location heading · MapKit / MapLibre offline pack · local notifications (SOS)

LEGEND

- LoRa
- NFC
- UWB (S2)
- BLE (OPT-07)
- GNSS
- dashed = Stretch 2 or optional

Links

LINK	BETWEEN	CARRIES	SECURITY	TIER
LoRa 915 MHz	Bracelet ↔ bracelet	24-byte packets every 2–10 s: position, time, battery, message type (position, SOS, quick message, meet-up, relayed)	AES-128-CCM with the group key; 4-byte MIC; dedupe by sequence number	MVP (relay: S2)
NFC	Bracelet ↔ bracelet, tap < 10 cm	Group key + device ID, once per pairing	Physical proximity is the consent step	MVP
GNSS L1	Satellites → each bracelet	Position, time, PPS pulse for TDMA slot timing	Receive only	MVP
UWB	Bracelet ↔ bracelet, ~30 m	Two-way ranging at 10 Hz in Finding mode	Ranging only, no user data	Stretch 2
BLE GATT	Bracelet (nRF52 bridge) ↔ iPhone	Peers (notify), Self status, Command (mode, message, meet-up, SOS, pair), Config	LE Secure Connections, 6-digit code on the bracelet; group key never sent	Optional (OPT-07)
Pogo dock / SWD	Bracelet ↔ dock / PC	Charging, firmware updates via the ST system bootloader, debug console; SWD for development	Physical access only; RDP level 1 on release builds	Stretch 1

490A prototype vs. final bracelet

PART OF THE SYSTEM	490A PROTOTYPE (DEV BOARDS)	490B FINAL BRACELET (CUSTOM PCB)
MCU board	NUCLEO-U5A5ZJ-Q ×3 (portable units); STM32U5G9J-DK1 ×1 for UI/GPU work	STM32U5G9VJT6Q on Rev A PCB
Display	GC9A01 1.28" round TFT (SPI) on the Nucleo units; DK1's 2.47" round TFT for UI development	Round AMOLED, CO5300 driver (QSPI) + touch
IMU	Adafruit BNO085 breakout on I2C	BNO085 on SPI
Haptics	ERM coin motor on a GPIO	DRV2605L + LRA
Input	Nucleo user button	PEC06 rotary crown + push
Power	USB power bank	500 mAh LiPo, BQ25180 charger, TPS63802, MAX17048, pogo dock
Radios	SX1262, MAX-M10S, PN7150 breakouts	Same chips as modules on the PCB; DWM3000 (S2), nRF52 bridge (OPT-07)
Debug	On-board ST-LINK	Tag-Connect TC2030 + STLINK-V3MINIE
Firmware	Same FreeRTOS application, services and drivers on both; only the board support package (pins, buses, clocks) changes	

Software on each device

DEVICE	SOFTWARE	LANGUAGE / TOOLS	LIVES IN REPO AT
Bracelet (STM32U5G9)	FreeRTOS firmware (7 tasks, tickless idle): app, LVGL UI, services, drivers, HAL/BSP	C11, STM32CubeMX / CubeIDE, FreeRTOS (X-CUBE-FREERTOS), LVGL 9	`firmware/bandlink/`
BLE bridge (nRF52, optional)	GATT server + UART framing to the STM32. Runs its own small event-driven loop with the BLE stack in the SoftDevice; it does not need FreeRTOS. Alternative: an AT-command BLE module (e.g. RN4871), so there is no bridge firmware to write.	C, nRF5 SDK + SoftDevice S112/S132	`firmware/ble_bridge/`

DEVICE	SOFTWARE	LANGUAGE / TOOLS	LIVES IN REPO AT
iPhone (optional)	BandLink app: radar, map, group, messages, settings	Swift, SwiftUI, Core Bluetooth, Core Location	`mobile/ios/`
Development PC	Flashing and log tools, PPK2 power analysis; host unit tests for services (planned)	C (host gcc), Python	`firmware/tests/`, `firmware/tools/`